

CIS241

System-Level Programming and Utilities

SSH

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Based on material provided by Erin Carrier and Austin Ferguson

First, Linux

What is it?

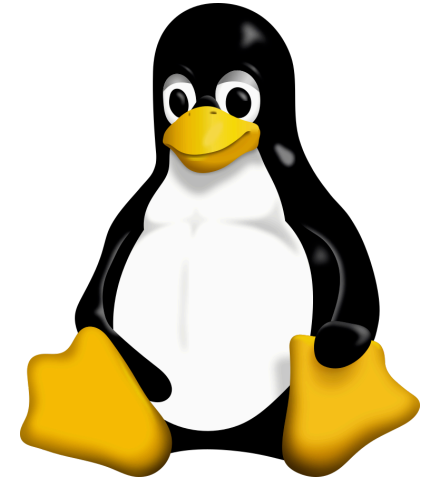
- Often referred to as an operating system (OS)
 - Competing with Mac/Windows
- *Technically*, it is a kernel

Many OSs use the Linux kernel

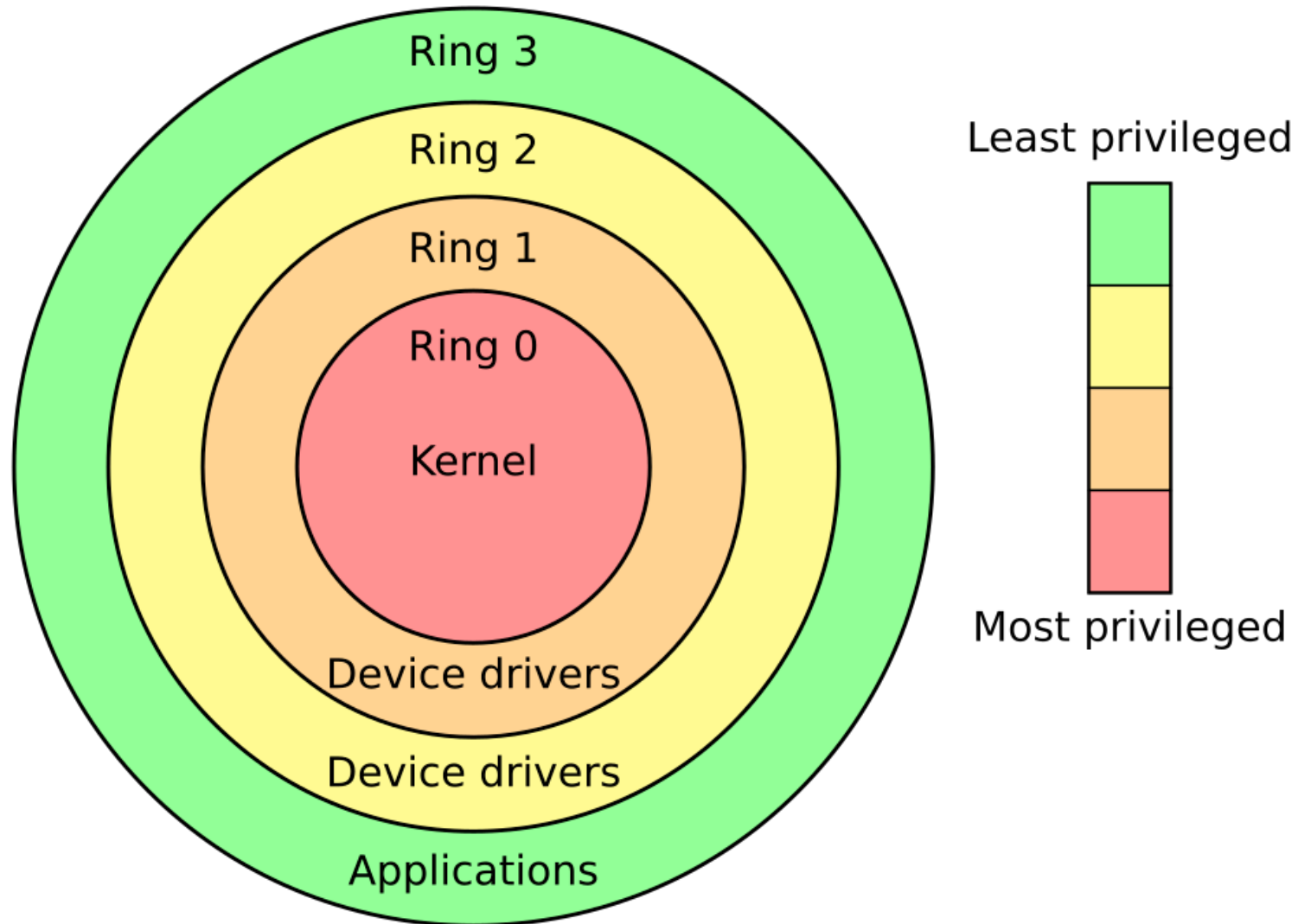
- Ubuntu (what is on EOS)
- Arch, Kali, Gentoo, Fedora, etc.

Isn't Mac Linux too?

- Nope, Darwin! (Unix-based)



Protection Rings



A bit of history

1970s: UNIX source code typically stripped from distribution

- Exorbitant amounts of money to acquire source

Linux first developed by Linus Torvalds to be a copy of UNIX

- Released it to Internet for suggestions on how to improve it
- Kernel only updated by core developers and from suggestions
- [Short history of Linux](#)



Unix?

Unix was an earlier operating system

- Linux and other systems are heavily influenced by Unix

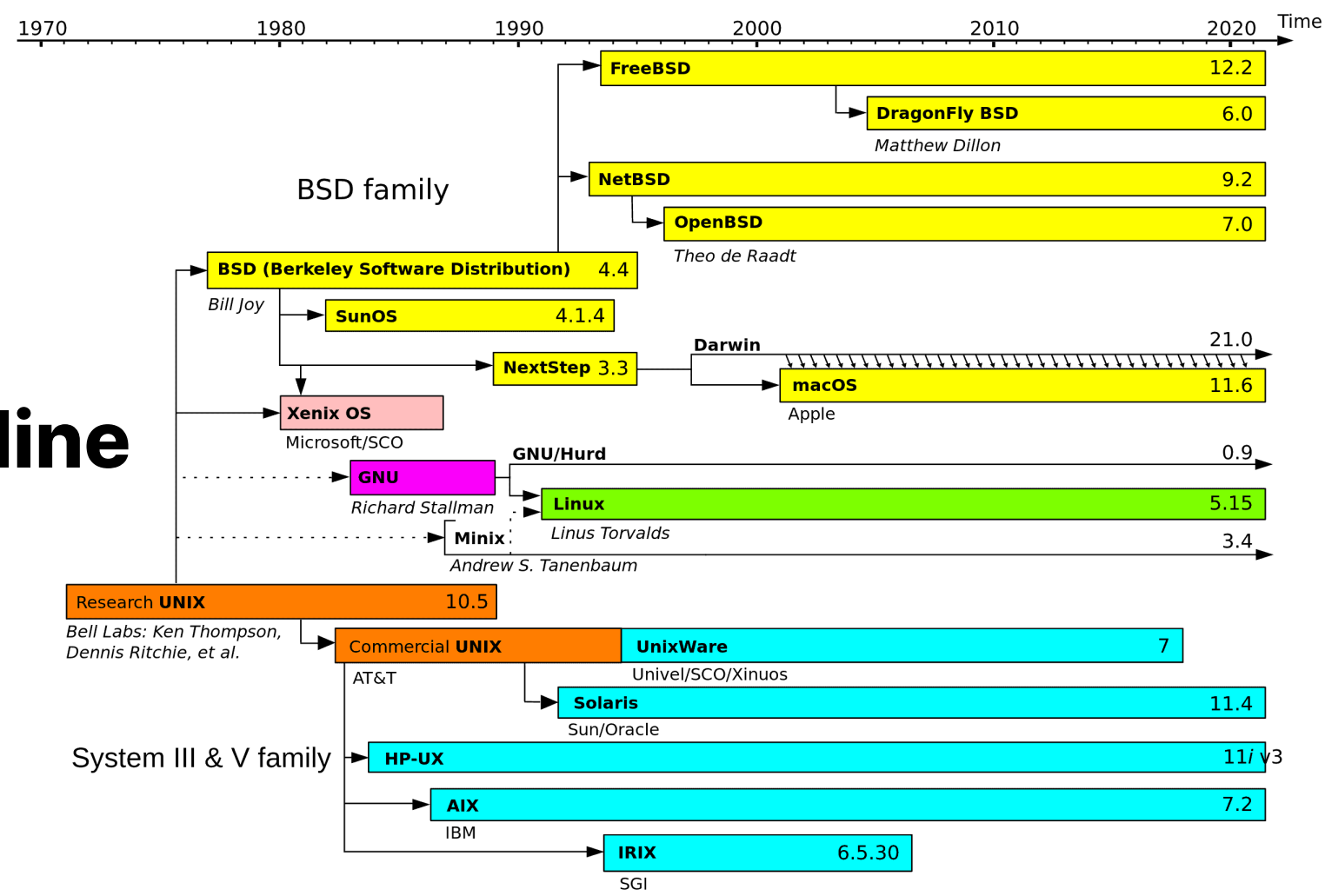
We now use "Unix-like" to describe systems

- They follow the same general ideas as Unix
- This means most things (commands, properties) carry over!

"Unix-like" systems:

- Linux
- macOS
- FreeBSD
- Solaris

Unix Timeline



Windows

Windows (today), is not Unix-like

- Can you do your homework on a Windows machine?
- Not by default

However, we have WSL! (🐧_🐧🌸)

- Windows Subsystem for Linux
- If you are on Windows, you must install WSL

<https://learn.microsoft.com/en-us/windows/wsl/install>

- Be very happy about this - I dreamed about just having the ability to SSH from the terminal when in industry

Terminal

```
erik@worktop: ~  
  
          ///////////  
        ///////////  
      //////////*767//////////  
    //////////76767676*/////////  
  //////////76767//76767/////////  
    //////////767676//*76767/////////  
  //////////767676//76767.//7676*/////////  
    //////////767676//76767//767676/////////  
  //////////76767676767//76767/////////  
    //////////76767676//7676/////////  
  //////////,7676,,//767/////////  
    //////////*7676////////76/////////  
  //////////7676////////767/////////  
    //////////7676//767/////////  
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    //////////.76767676767676767,/////////  
  //////////7676767676767676767/////////  
    ///////////  
  ///////////  
  ///////////  
  
erik@worktop  
-----  
OS: Pop!_OS 22.04 LTS x86_64  
Host: Latitude 7400  
Kernel: Linux 6.12.10-76061203-generc  
Uptime: 2 days, 34 mins  
Packages: 4267 (dpkg), 7 (flatpak-us)  
Shell: bash 5.1.16  
Display (AU0243D): 1920x1080 @ 60 Hz]  
DE: GNOME 42.9  
WM: Mutter (X11)  
WM Theme: Pop-dark  
Theme: Pop-dark [GTK2/3/4]  
Icons: Pop [GTK2/3/4]  
Font: Fira Sans (10pt, SemiLight) [G]  
Cursor: Pop (24px)  
Terminal: GNOME Terminal 3.44.0  
Terminal Font: Fira Mono (12pt)  
CPU: Intel(R) Core(TM) i7-8665U (8) z  
GPU: Intel UHD Graphics 620 @ 1.15 G]  
Memory: 15.70 GiB / 31.16 GiB (50%)  
Swap: 0 B / 20.00 GiB (0%)  
Disk (/): 421.50 GiB / 928.94 GiB (44  
Disk (/recovery): 2.84 GiB / 3.99 GiB  
Local IP (wlo1): 192.168.1.46/24  
Battery (DELL 5VC2M85): 100% [AC Con]  
Locale: en_US.UTF-8  
  
erik@worktop:~$
```


Shells (local)

See the little dollar sign?

Shell!

What shell are we using?

```
echo $0
```



The screenshot shows a terminal window titled "erik@worktop: -". On the left, there is a large ASCII art logo of a triangle composed of the characters "767". On the right, the terminal displays the following system information:

```
erik@worktop
-----
OS: Pop!_OS 22.04 LTS x86_64
Host: Latitude 7400
Kernel: Linux 6.12.10-76061203-generc
Uptime: 2 days, 34 mins
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```

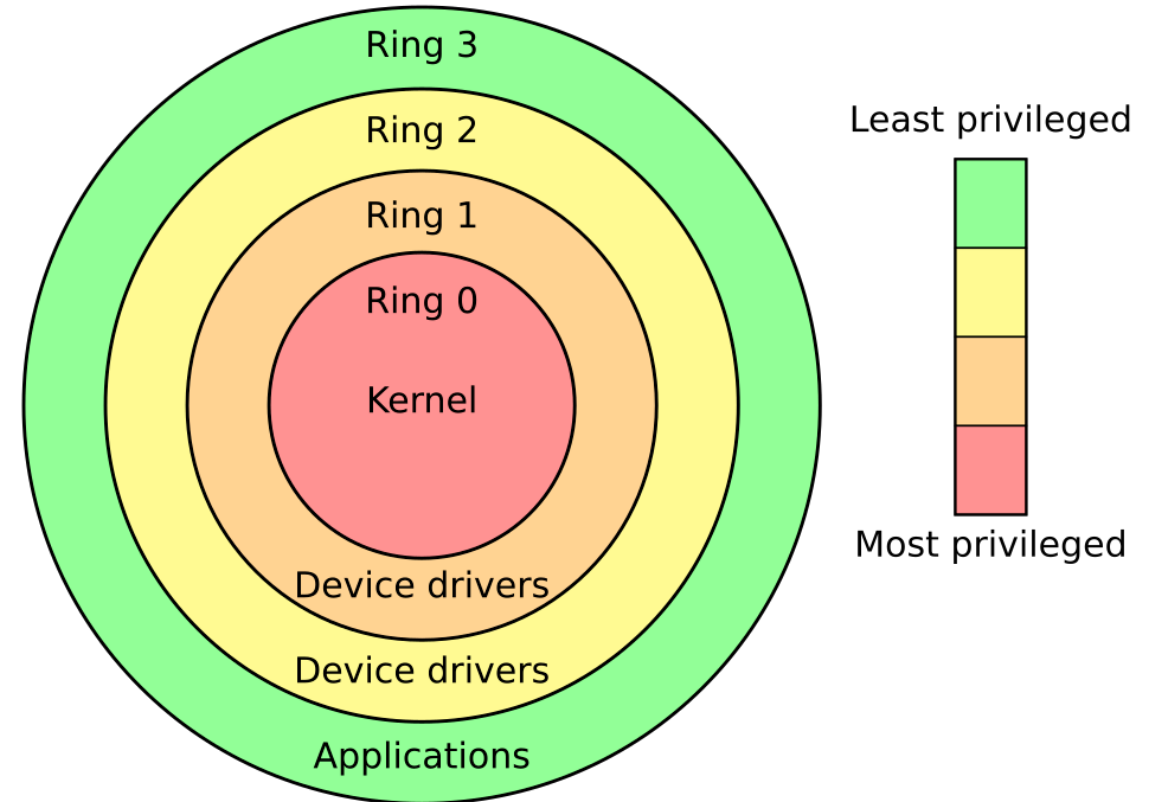
At the bottom left, the prompt "erik@worktop:~\$" is visible, followed by a cursor. At the bottom right, there is a small color calibration bar.

Shells (local)

Where is the shell here?

- Ring 3, typically!

How do we run in other rings?



You can use `eos01.cis.gvsu.edu` - `eos30.cis.gvsu.edu` !

Note - if you're off-campus you'll need a VPN!

Shells (remote)

Or, connecting to **another** machine

- Or or, doing your homework ๖_๖

`ssh` → **Secure Shell**

Usage: `ssh username@address`

For example, connecting to EOS:

```
ssh username@eos1.cis.gvsu.edu
```

Lab fun

DO NOT

SHUT DOWN

THESE MACHINES

